# Text as Pieces

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

## Teach tokenization as a specified representation procedure.

Supply complete vocabularies and programs, separate the cases and preserve the fresh phrase assessment.

**Keep one habit:** Require original text, IDs, space handling, counting convention and failure position beside any token count.

MAIN LESSON · SECTION 1

## Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

### Learning goal and precision

Primary target: Encode and decode text under an explicit toy **vocabulary**

Use two 45-minute sessions. All model **output**s and probability tables are constructed classroom examples. No hosted AI service, external API or student account is needed.

Distinguish a model objective from factual support and system permissions. Collect independent work before discussing its solutions.

By the end, I can encode and decode text using a specified **vocabulary** and matching rule.

## What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Printed lesson and workbook; paper and pencil
- Calculator; optional local Python 3 interpreter
- Prepare the independent task and fresh retry without displaying solutions

## Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

## Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Feedback Loops. Give an example and a mistake that the idea helps you avoid.

Look for: Implement a deterministic recommender

Reference example, not the only acceptable wording: Both score 8 and qualify; A is alphabetically first under the tie rule.

Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

Look for: Compute absolute distances and rank **training** neighbours

Reference example, not the only acceptable wording: Distances to A,B,C,D,E are 1,1,5,7,9 cm. A and B are equally far from 3 on the numeric feature; their labels do not enter the distance calculation.

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

Look for: Define a question and target population

Reference example, not the only acceptable wording: Population: all Grade 7 students. Only lunch-club members or one friend group is an unsuitable basis for a whole-grade claim.

An exact representation must reconstruct the original source. Lowercasing, removing punctuation or changing spaces is a transformation and must be declared separately rather than called an exact round trip.

▼ Earlier wording, alternatives and picture transcripts

### Teach the learning cycle

### Retrieval answer guidance

### Feedback Loops

### Nearest Neighbor Reasoning

### Sampling and Representation

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### What the learner must demonstrate

Toy **token**izer A splits 'unhelpful' into un/help/ful; B keeps it whole. Count **token**s for each and explain why a three-word sentence need not use three **token**s.

### Concept to explain

Language systems **process token**s and **pattern**s rather than whole human meanings.

| Term | Meaning |
|---|---|
| token | A unit of encoded text processed by a language model, which may be a word, part of a word or punctuation. |
| tokenization | Converting text into the token units of a particular tokenizer. |
| context | Information available around a task that helps interpret or answer it. |
| pattern | A relationship or regularity observed in examples. |

### Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

### Preparation for the connected explanation

### Readiness answer

### End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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MAIN LESSON · SECTION 2

# Teach a tokenizer through an explicit vocabulary and trace

> Keep the cat and book vocabularies distinct.

## State vocabulary, matching and unsupported-input policy

**Token**s need not be words. A classroom **token**izer can use an exact, case-sensitive **vocabulary** and greedily choose the longest matching **token** from left to right. This is a toy rule, not a claim about every production **token**izer.

Define IDs: 0=end, 1=cat, 2=c, 3=a, 4=t, 5=s, 6=space, 7=sat. The space **token** represents one literal space. The end **token** is a special terminator, not ordinary text.

At each position choose the longest matching ordinary **token**. If no **token** matches, stop and report the unsupported character and its position. Do not delete the character or guess a replacement.

IDs identify entries; a larger ID does not mean a **token** is more valuable or more similar.

## Explain exact decoding and token-count conventions

The **encoding** of cats sat is [1,5,6,7], followed by [0] only if the protocol appends an end **token**. **Decoding** concatenates the ordinary **token** strings to recover cats sat.

Count four text **token**s, or five sequence items with the end marker included; always state the counting convention. The two written words do not imply two **token**s.

## Compare original and character-only cat representation

A character-only **vocabulary** c,a,t,s,space could encode cats sat as c,a,t,s,space,s,a,t: eight text **token**s. The original **vocabulary** uses four. Both decode to the same source text.

Sequence length is a property of the text, **vocabulary** and **encoding** rule together. Counts from different **token**izers are not automatically comparable.

In this **vocabulary**, sat is chosen over s because it is longer. The toy longest-match rule is defined explicitly. Real **token**izers may use different algorithms, byte **encoding**s and special-**token** policies.

Never infer an unknown **token**izer's count from word count alone.

## Run the complete supported/failing input audit

```
vocabulary = {1: "cat", 2: "c", 3: "a", 4: "t", 5: "s", 6: " ", 7: "sat"}
def encode(text, entries):
    ids = []
    position = 0
    order = sorted(entries, key=lambda number: (-len(entries[number]), number))
    while position < len(text):
        found = None
        for token_id in order:
            if text.startswith(entries[token_id], position):
                found = token_id
                break
        if found is None:
            return ids, (position, text[position])
        ids.append(found)
        position += len(entries[found])
    return ids, None
for text in ["cats sat", "c at", "Cat", "cat!"]:
    ids, error = encode(text, vocabulary)
    print(repr(text), ids, error)
```

```
    if error is None:
        decoded = "".join(vocabulary[token_id] for token_id in ids)
        print("decoded", repr(decoded), "text tokens", len(ids))
character_entries = {number: text for number, text in vocabulary.items() if
len(text) == 1}
ids, error = encode("cats sat", character_entries)
print("character-only", ids, len(ids), error)
```

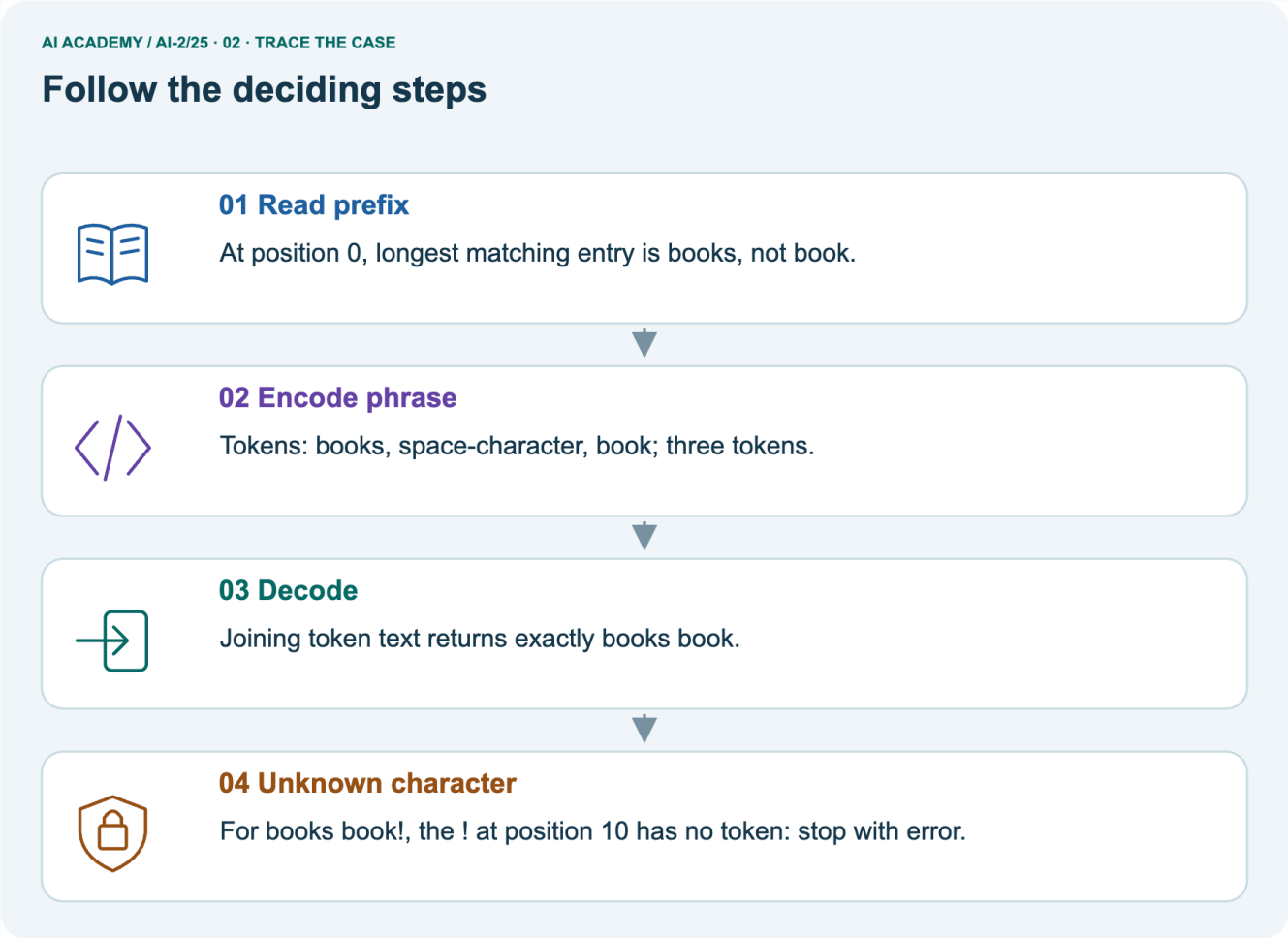
## Introduce the separate book vocabulary and phrase

A fictional **token**izer converts a short catalogue phrase into **token**s.

**Vocabulary** entries: book, books and the space character. At each position choose the longest matching entry; preserve the actual space.

Text is “books book”. Exclamation marks have no **token**; zero-based character positions are used.

## Trace its exact text and unknown character




Follow the labelled picture with the matching explanation. Open the vector diagram

A **token** is defined by the **vocabulary** and **encoding** rule; it need not be one word or one character. Longest matching matters where **vocabulary** entries overlap. Counting text **token**s here does not count letters.

Preserve spaces during **decoding** so the reconstructed text matches the original exactly.

This deterministic toy **token**izer omits real subword **training**, Unicode handling and model-specific special **token**s.

## Change only the vocabulary

Change: Use a character **vocabulary** covering b,o,k,s and space instead.

**Prediction**: The same phrase uses ten character **token**s.

Reason: There are five letters, one space and four letters; its decoded text remains unchanged.

## Elicit picture-based explanation

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

### ▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

**Input** → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. ‘unhelpful’ might split as ‘un’ + ‘help’ + ‘ful’.

2. A name or emoji may use unexpected **token** pieces.

3. **Token** boundaries need not match whole words.

4. Meaning emerges from **pattern**s across **context**, not from one **token** acting like a dictionary.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

#### Concrete teaching case

Toy **token**izer A splits rainbows as [rain][bow][s]; B uses [rainbows].

Count **token**s and state the limit of the example.

### Exact explanation to model

Three versus one. **Token** count depends on **token**izer; **token**s need not equal words or letters.

### Explain the added worked example

### Key terms

| Term | Meaning |
|---|---|
| Token | An item in a model vocabulary, which may represent a word, part of a word or another text unit |
| Vocabulary | The permitted tokens and their identifiers |
| Encoding | Mapping source text into a token sequence under a specified rule |
| Decoding | Mapping a token sequence back to text under a specified rule |

### Model the reasoning aloud

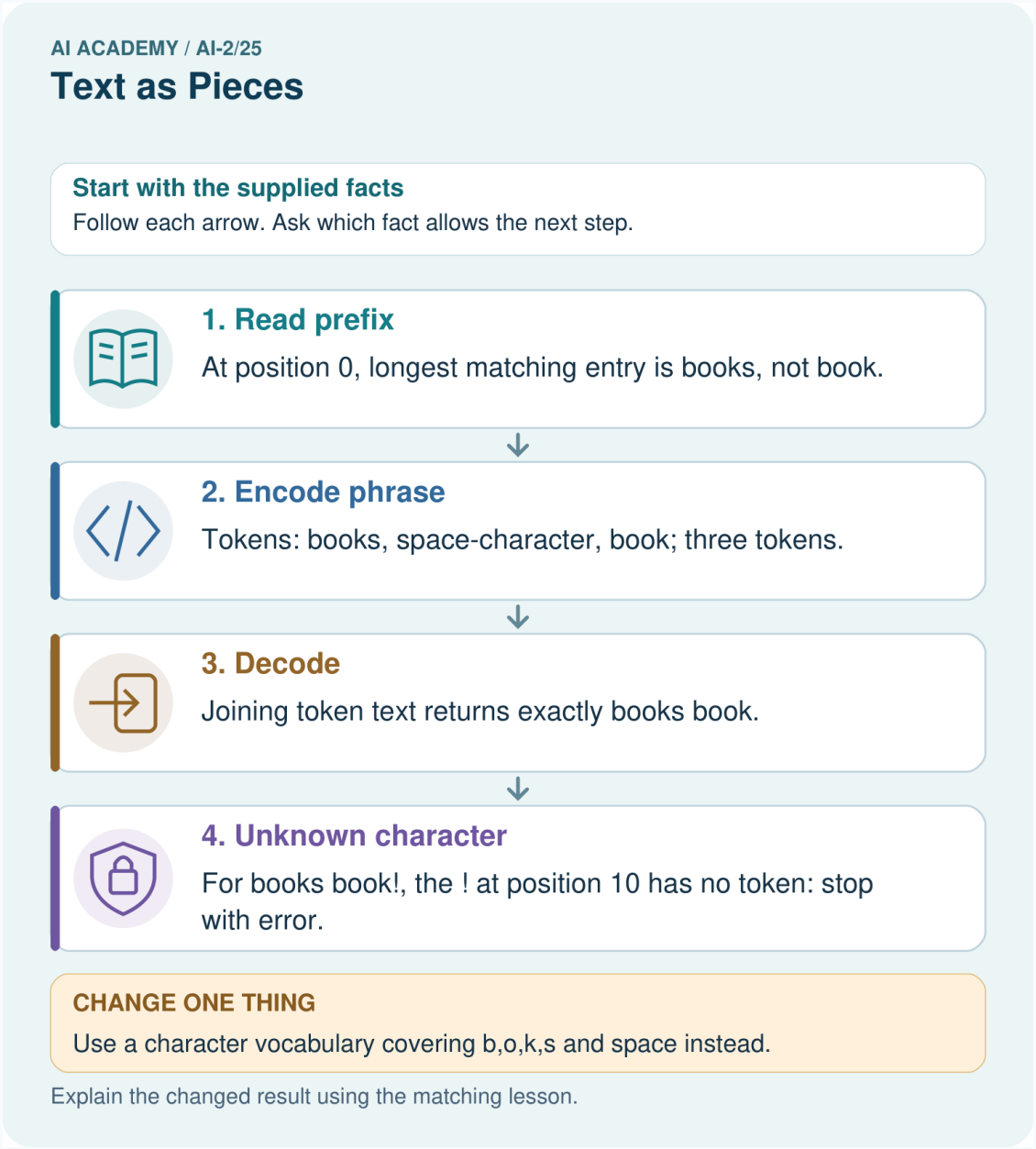
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

| Thinking move | Teacher prompt |
|---|---|
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

## A pictorial worked case: Text as Pieces

### The facts we will use




Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Read prefix | At position 0, longest matching entry is books, not book. |
| 2. Encode phrase | Tokens: books, space-character, book; three tokens. |
| 3. Decode | Joining token text returns exactly books book. |
| 4. Unknown character | For books book!, the ! at position 10 has no token: stop with error. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

### What this example does not establish

### Change one thing in the pictured case

### Ask while showing the picture

# Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

## Prompt before revealing

Completion task: Encode cats sat using the given **vocabulary**, decode it, and count with and without an appended end marker.

## Expected reasoning and fading help

Teacher check: Teacher explanation: Text IDs [1,5,6,7] decode cats sat. Four text **token**s; five sequence items if the protocol appends end ID 0.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

## Check the misconception before discussion

Use the worked case for **Token**isation and **vocabulary**. Proposed explanation: “A **token** always equals a word”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The **vocabulary** contains both whole strings and character pieces, so one word can take multiple **token**s.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

## Use the pictorial check

Why is the unknown ! at position 10 rather than position 9?

books occupies 0–4, space is 5, book is 6–9, so ! is character 10.

Changed-case reasoning: The same phrase uses ten character **token**s. There are five letters, one space and four letters; its decoded text remains unchanged.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

## Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

### ▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

#### Board sequence

Write the given facts: Toy **token**izer A splits rainbows as [rain][bow][s]; B uses [rainbows].

Write the question to explain: Count **token**s and state the limit of the example.

Reveal after **prediction**s: Three versus one. **Token** count depends on **token**izer; **token**s need not equal words or letters.

Use the earlier changed question as optional later practice. Reserve Trace a new input independently for the first independent attempt before hints or feedback.

## Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

## Model these guided questions

Guided: c at gives [2,6,3,4], four text **token**s. cat! fails on ! at zero-based position 3 after partial [1].

Exit: deleting ! changes the **input**; report unsupported punctuation or use a separately specified pre**process**ing/versioned **vocabulary**.

## Case-specific teaching sequence

| Stage | Teacher action |
| --- | --- |
| Session 1 start to 7 minutes | Retrieve the prerequisite and distinguish a numerical claim from its interpretation. |
| 7 to 18 minutes | Read See how it works and Words that unlock the idea, then complete W1; ask for a reason alongside each answer. |
| 18 to 32 minutes | Trace Follow a worked example and complete W2. |
| 32 to 45 minutes | Discuss the diagram and complete W3 in pairs; record support. |
| Session 2 start to 8 minutes | Complete the guided response before sharing the answer. |
| 8 to 20 minutes | Read the remaining sections and trace another constructed case. |
| 20 to 35 minutes | Collect W4 independently, requiring calculations and a bounded conclusion. |
| 35 to 45 minutes | Discuss W5 and the exit response. Provide feedback and a fresh retry. |

## Reduce help deliberately

## Individual reasoning check

## Pictorial practice question

## Reasoned teacher answer

MAIN LESSON · SECTION 4

# Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

### Supplied materials and answer guidance

cats sat→[1,5,6,7],exact decode,four ordinary tokens,five with appended end. Character-only→[2,3,4,5,6,5,3,4],eight ordinary tokens. c at→[2,6,3,4]. Cat fails uppercaseC at0; cat! retains[1] then fails! at3. Separate books book has three chunk tokens or ten characters; ! is at10. Token IDs are not semantic distances, and no production tokenizer or continuation model is implemented. Reserve the fresh phrase pair for W4.

## Run and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Use the complete supplied toy encoder and four exact source strings. Record ordinary IDs, counting convention, decoded text or partial failure, then compare character-only vocabulary. No continuation model is supplied, so do not invent a probability change after removing **context**.
4. What did the hardest test teach you that the normal test did not?

## Keep an inspectable record

| Record | What to capture |
| --- | --- |
| Input or evidence | Exact data, instruction or claim |
| Expected result | Answer or safe fallback declared before the run |
| Observed result | Actual result, including errors |
| Explanation | Rule, calculation or evidence explaining the outcome |
| Next test | One justified change and a new prediction |

## Diagnose the first misunderstanding

A **token** always equals a word: The **vocabulary** contains both whole strings and character pieces, so one word can take multiple **token**s.

IDs measure semantic similarity: IDs are **vocabulary** indices; their numerical spacing does not establish meaning.

Dropping characters preserves an exact round trip: Deleted characters change the source and must not be hidden.

## Choose support from the response

| What the response shows | Next teaching action |
| --- | --- |
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |

| What the response shows | Next teaching action |
| --- | --- |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

### ▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

#### Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Trace a new input independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

#### Diagnose and reteach

For guessing, return to given facts and request a trace. For **vocabulary** without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

#### Address these additional misconceptions

#### Match the support to the first error

MAIN LESSON · SECTION 5

# Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

## Selected independent question and key

Selected case: Under the original **vocabulary** encode sat cats, decode it and count text **token**s. Then trace sat cat! and state the exact failure position using zero-based character positions.

Teacher explanation: sat cats encodes [7,6,1,5], decodes exactly and has four text **token**s. sat cat! has partial IDs [7,6,1] then unsupported ! at position 7. Report failure with the partial trace, not complete success.

## Judge explanation, trace and limits

| Dimension | Evidence required |
|---|---|
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

## Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

### W1 Read the contract

Explain why sat wins over s at the start of sat and why uppercase C is unsupported.

Teacher explanation: The rule chooses the longest matching ordinary **token**; sat has length three and s length one. **Vocabulary** and matching are case sensitive and contain no uppercase C.

### W2 Encode and decode

Encode cats sat using the given **vocabulary**, decode it, and count with and without an appended end marker.

Teacher explanation: Text IDs [1,5,6,7] decode cats sat. Four text **token**s; five sequence items if the protocol appends end ID 0.

### W3 Compare sequence lengths

Encode cats sat with character **token**s only. Explain the difference from four text **token**s in the original **vocabulary**.

Teacher explanation: c,a,t,s,space,s,a,t is eight text **token**s. Larger chunks in the original **vocabulary** compress this representation; the source text is unchanged.

## W5 Audit silent repair

An encoder lowercases Cat and removes punctuation before **encoding**, but labels its **output** an exact **encoding** of the original. Explain what must be recorded or changed.

Teacher explanation: Lowercasing and punctuation removal alter the source.

Record pre**process**ing separately and label the transformed text; an exact encoder under this contract must report unsupported characters rather than silently change them.

### ▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

#### Worked case answer

Three versus one. **Token** count depends on **token**izer; **token**s need not equal words or letters.

#### Additional transfer practice question

A classroom **token**izer splits sunflower into sun and flower. How many pieces result? Must every real **token**izer split it the same way?

#### Reasoned practice answer

Two pieces result under this supplied scheme. Real **token**izers use different vocabularies and rules, so their splits may differ.

#### What counts as evidence

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Result | Incorrect or absent | Partly correct | Correct with necessary conditions |
| Reasoning | No supporting trace | Some steps | Clear mechanism, calculation or source support |
| Limit | Overclaims certainty | Vague limitation | Relevant uncertainty or missing fact |

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Transfer | Copies without adapting | Adapts with prompts | Solves a changed case independently |

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

### Feedback

Ask the learner to find the **evidence** that challenges this claim: “One word always equals one **token**.” Keep the first attempt and an uncoached retry separate.

### Original transfer answer

A uses 3 **token**s, B uses 1. Real **token**izers can split words, include spaces or punctuation and use different vocabularies.

A word count alone cannot determine a **token** budget; these toy splits illustrate the idea rather than document a particular model.

### Explanations for the additional workbook examples

### W1 Read the contract

### W2 Encode and decode

### W3 Compare sequence lengths

### W4 Trace a new input independently

Under the original **vocabulary** encode sat cats, decode it and count text **token**s. Then trace sat cat! and state the exact failure position using zero-based character positions.

Teacher explanation: sat cats encodes [7,6,1,5], decodes exactly and has four text **token**s. sat cat! has partial IDs [7,6,1] then unsupported ! at position 7. Report failure with the partial trace, not complete success.

**W5 Audit silent repair**

**Assess independent transfer**

**Selected case answer**

**Feedback that the learner uses**

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MAIN LESSON · SECTION 6

# Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

## Project evidence to keep

**Token**ize book descriptions before any text-based matching, while preserving the original source text.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Document the recommender's text representation and unknown-**token** behavior rather than assuming every model counts words identically.

## Support while preserving the reasoning goal

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Provide labelled intermediate columns and allow a calculator. Ask students to state the meaning and denominator of a value before interpreting it.

Extension: implement the classroom calculation locally and test boundary cases without using personal data.

W4 requires [7,6,1,5], four text **token**s and unsupported ! at position 7. Fresh retry: cats cat encodes [1,5,6,1], four text **token**s; cats Cat fails at uppercase C position 5 after [1,5,6].

Require exact **decoding** and an explicit failure.

## Plan the delayed check

Suggested return points in this level: ai-2/26, ai-2/28, ai-2/31. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

## Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 26: Image and Voice Pipelines. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week26: Image and Voice Pipelines. Carry forward explicit representation, preprocessing and failure records when tracing another modality; do not describe this week as a real model's **token**izer.

## ▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

### Transparent Book Recommender

### Support and stretch

### Exit reflection

### Bridge to the next week

### Support and fresh reassessment

### Plan the delayed return

### Project evidence from this case

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

## Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

### Preparation and schedule

Teaching target: Encode and decode text under an explicit toy **vocabulary**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

## Practice and assessment reminders

Before class, solve the supported practice: Encode cats sat using the given **vocabulary**, decode it, and count with and without an appended end marker.

Reasoned practice answer: Teacher explanation: Text IDs [1,5,6,7] decode cats sat. Four text **token**s; five sequence items if the protocol appends end ID 0.

Check every learner before discussion: Use the worked case for **Token**isation and **vocabulary**. Proposed explanation: “A **token** always equals a word”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The **vocabulary** contains both whole strings and character pieces, so one word can take multiple **token**s.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Use a character **vocabulary** covering b,o,k,s and space instead. Require a **prediction** and a reason before comparing. Expected result: The same phrase uses ten character **token**s.

Reason: There are five letters, one space and four letters; its decoded text remains unchanged.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

# Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

## Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

### Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

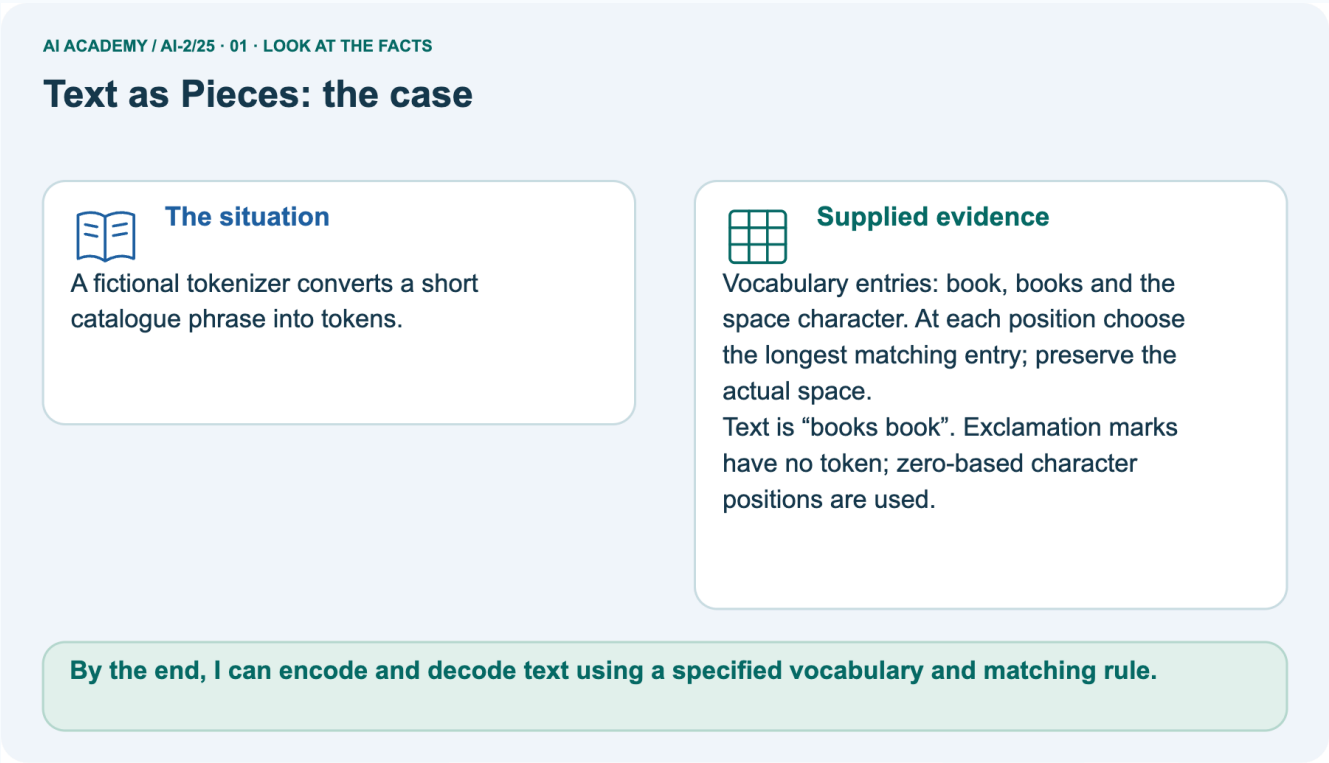
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

## Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

## 01. Text as Pieces: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

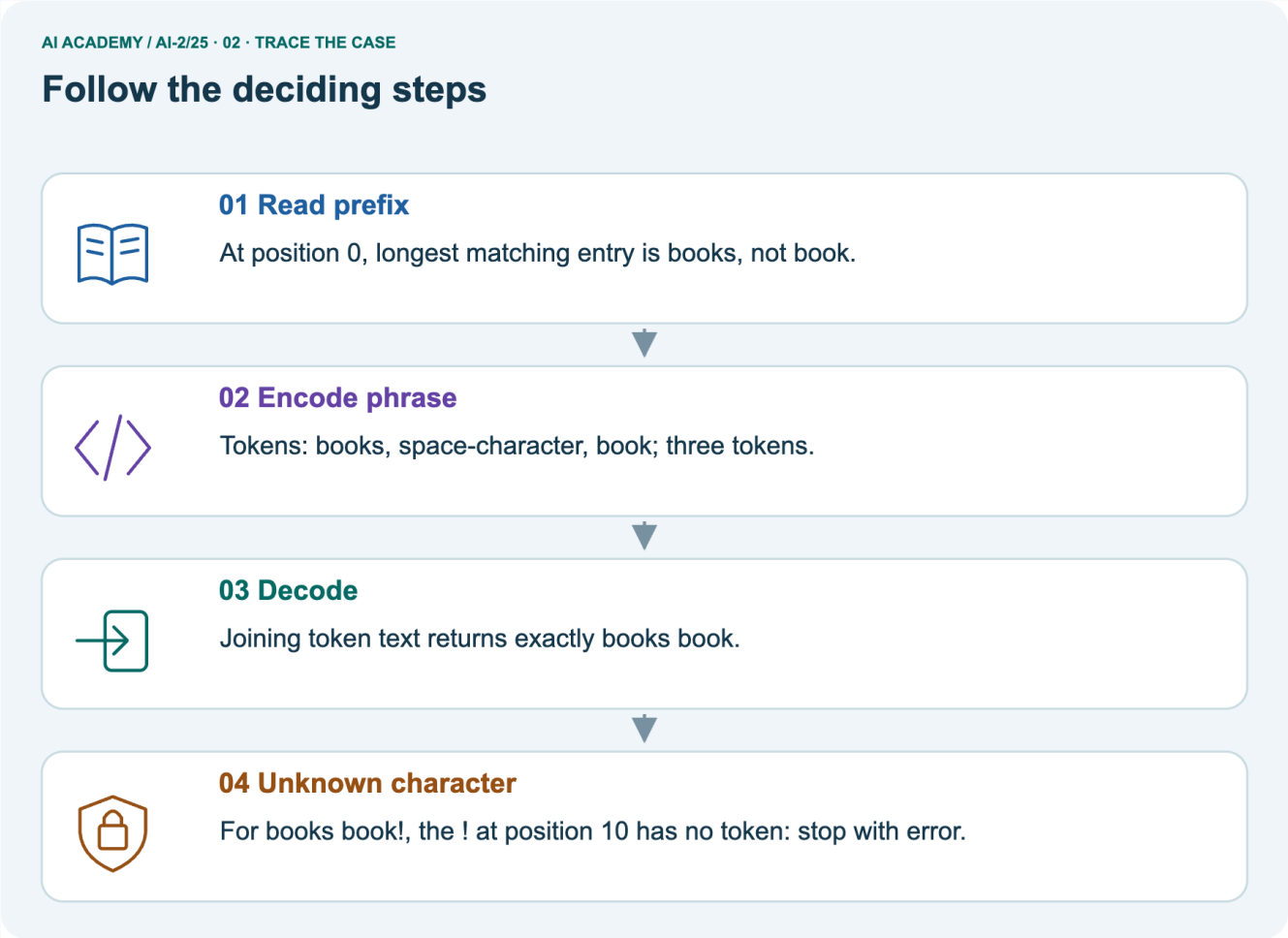
A fictional **token**izer converts a short catalogue phrase into **token**s. Supplied facts: **Vocabulary** entries: book, books and the space character. At each position choose the longest matching entry; preserve the actual space.

Text is “books book”. Exclamation marks have no **token**; zero-based character positions are used.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps



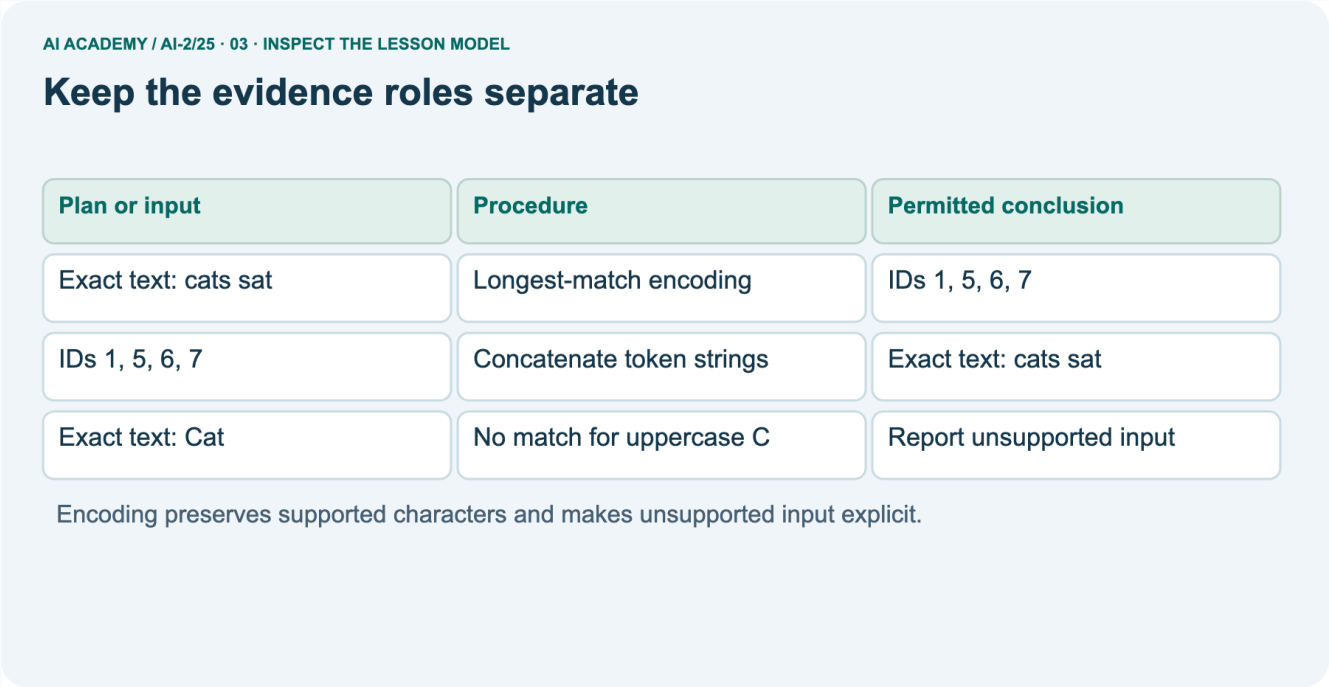

Follow the labelled picture with the matching explanation. Open the vector diagram

Read prefix: At position 0, longest matching entry is books, not book. → Encode phrase: **Token**s: books, space-character, book; three **token**s. → Decode: Joining **token** text returns exactly books book. → Unknown character: For books book!, the ! at position 10 has no **token**: stop with error.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 03. Keep the evidence roles separate




Follow the labelled picture with the matching explanation. Open the vector diagram

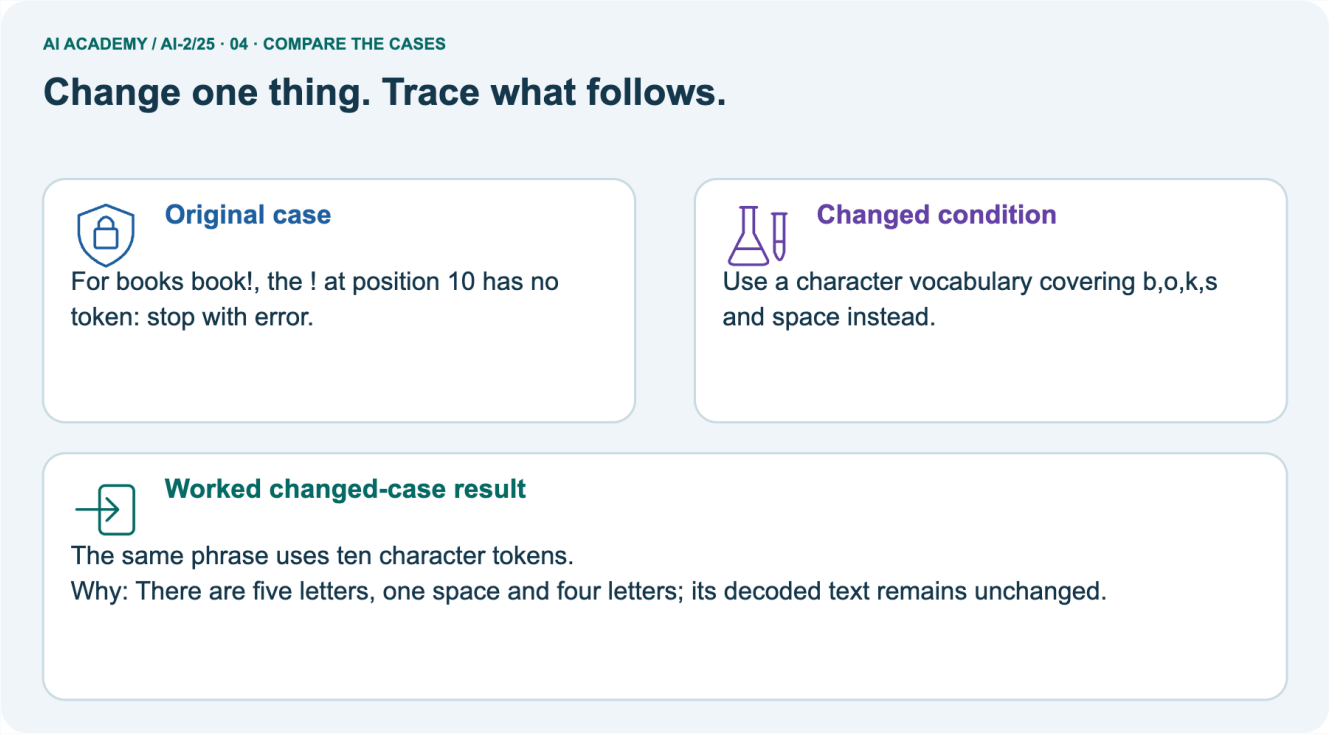
Keep the **evidence** roles separate. Exact text: cats sat | Longest-match **encoding** | IDs 1, 5, 6, 7. IDs 1, 5, 6, 7 | Concatenate **token** strings | Exact text: cats sat.

Exact text: Cat | No match for uppercase C | Report unsupported **input**

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: For books book!, the ! at position 10 has no **token**: stop with error. Changed condition: Use a character **vocabulary** covering b,o,k,s and space instead. Expected result: The same phrase uses ten character **token**s.

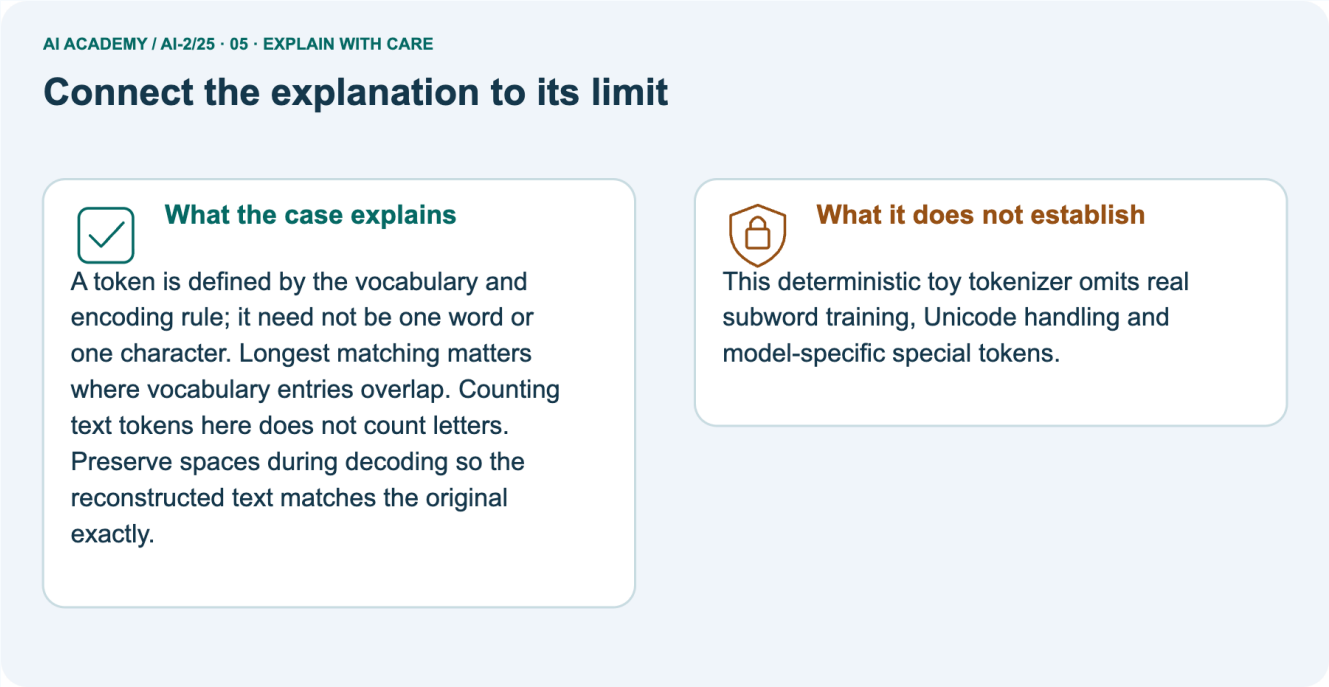
Reason: There are five letters, one space and four letters; its decoded text remains unchanged.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: There are five letters, one space and four letters; its decoded text remains unchanged.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

A **token** is defined by the **vocabulary** and **encoding** rule; it need not be one word or one character. Longest matching matters where **vocabulary** entries overlap. Counting text **token**s here does not count letters.

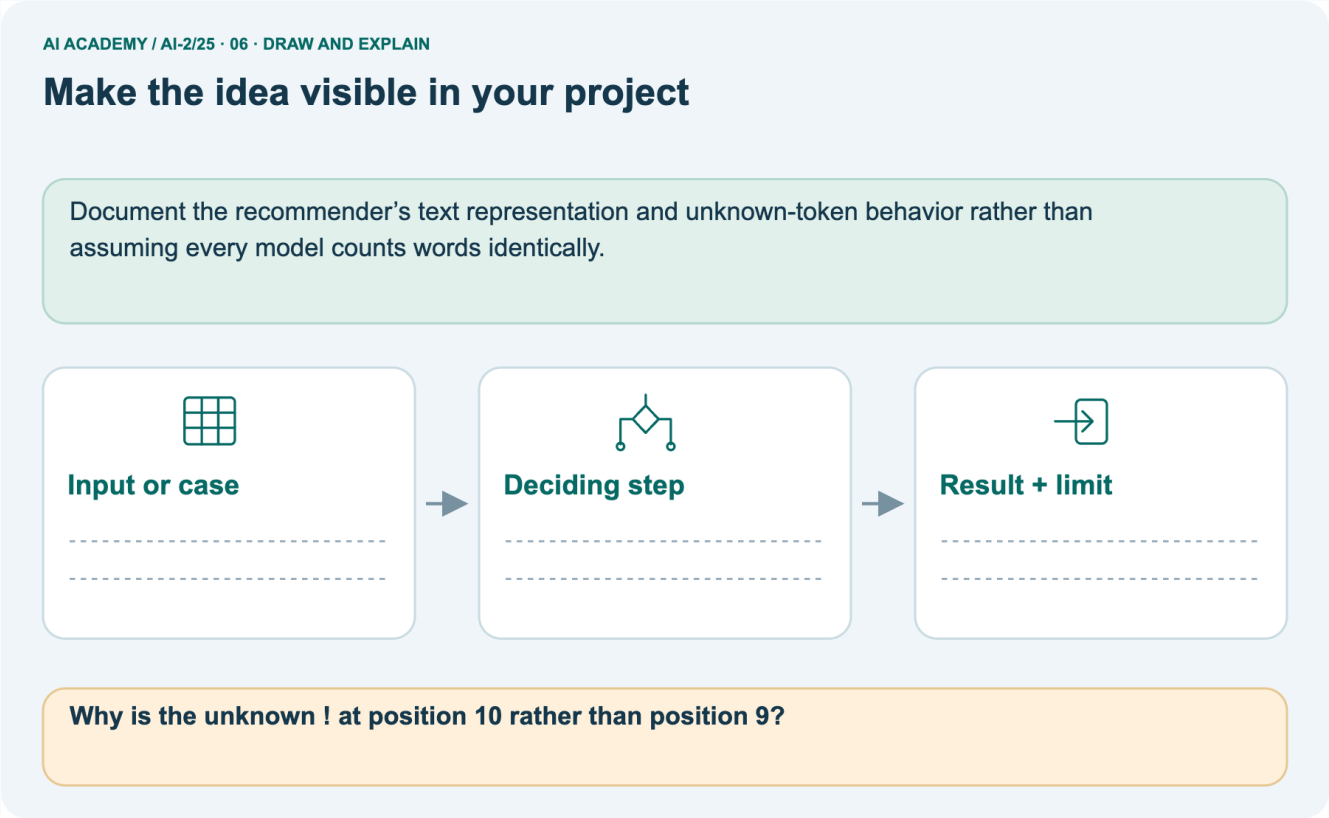
Preserve spaces during **decoding** so the reconstructed text matches the original exactly. Limit: This deterministic toy **token**izer omits real subword **training**, Unicode handling and model-specific special **token**s.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This deterministic toy **token**izer omits real subword **training**, Unicode handling and model-specific special **token**s.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Document the recommender’s text representation and unknown-**token** behavior rather than assuming every model counts words identically. Draw your **input**, deciding step, result and limit.

Supported question: Why is the unknown ! at position 10 rather than position 9?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: books occupies 0–4, space is 5, book is 6–9, so ! is character 10.

## ▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

### 01. Text as Pieces: the case

### 02. Follow the deciding steps

### 03. Keep the evidence roles separate

### 04. Change one thing. Trace what follows.

### 05. Connect the explanation to its limit

### 06. Make the idea visible in your project

## ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.